

Project Title : Blood Group Detection using Fingerprint using Machine Learning
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ABSTRACT

Blood group identification is an important requirement in medical emergencies. Traditional methods require blood samples and laboratory testing, which may take time. This project proposes a non-invasive approach to predict blood groups using fingerprint patterns. The system analyzes fingerprint features and applies machine learning techniques to classify blood groups such as A, B, AB, and O. This method aims to provide a fast, cost-effective, and contactless solution for preliminary blood group identification, especially useful in emergency situations.

INTRODUCTION

Blood group detection plays a vital role in healthcare systems, especially during accidents, surgeries, and emergency treatments. Conventional blood testing methods require physical blood samples and laboratory equipment. To overcome this limitation, this project introduces a system that uses fingerprint patterns to predict blood groups using machine learning techniques. Since fingerprints are unique and easily accessible, they can be used as a biometric input for prediction systems.

OBJECTIVES

- To develop a non-invasive blood group prediction system
- To use fingerprint patterns for classification
- To apply machine learning algorithms for accurate prediction
- To reduce time and dependency on lab testing
- To provide quick emergency identification support

METHODOLOGY

- Collect fingerprint dataset
- Preprocess fingerprint images (noise removal, enhancement)
- Extract features from fingerprint patterns
- Train machine learning model
- Test model with new fingerprint inputs
- Predict blood group classification

Technologies used:

- Python
- Machine Learning (Scikit-learn)
- OpenCV
- NumPy & Pandas

MODULES

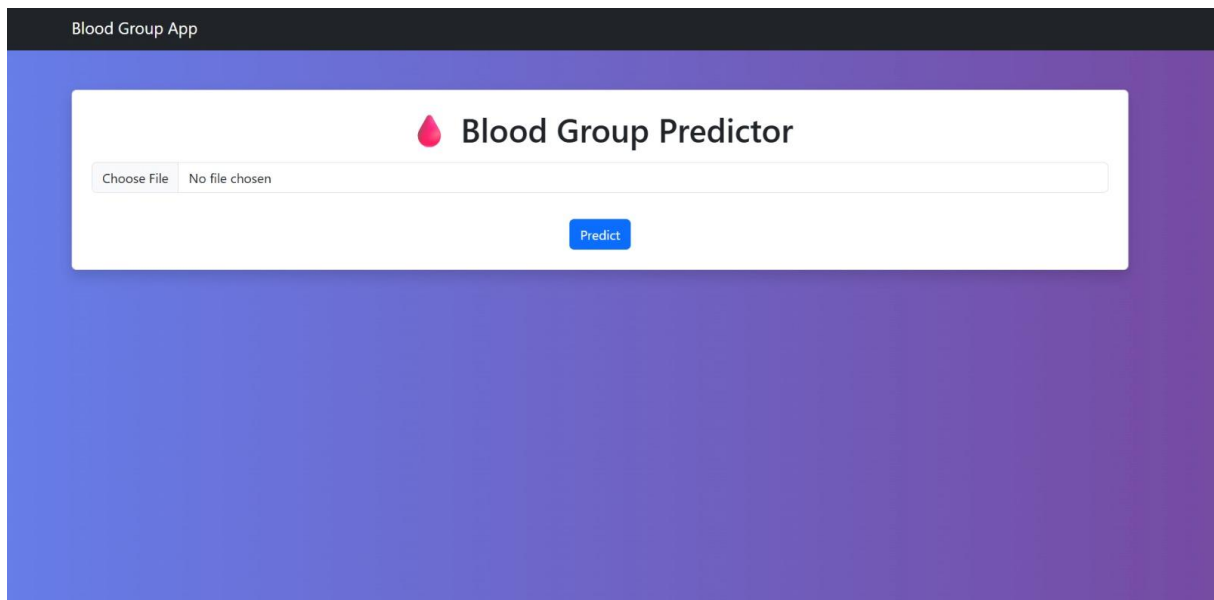
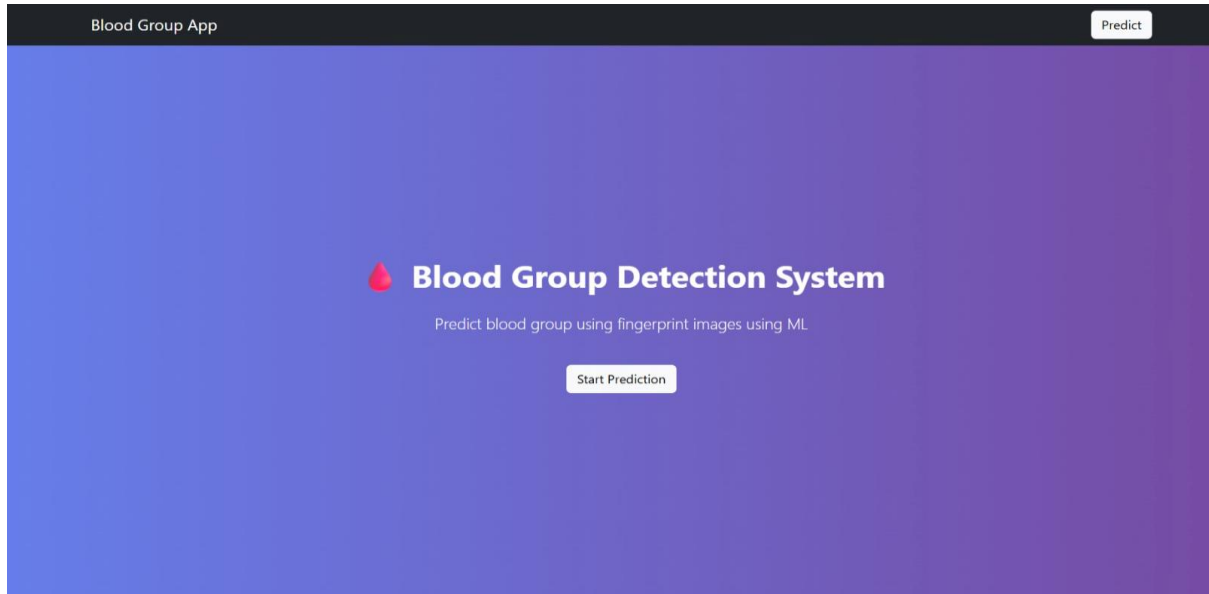
- Data Collection Module
- Image Preprocessing Module
- Feature Extraction Module
- Model Training Module
- Prediction Module

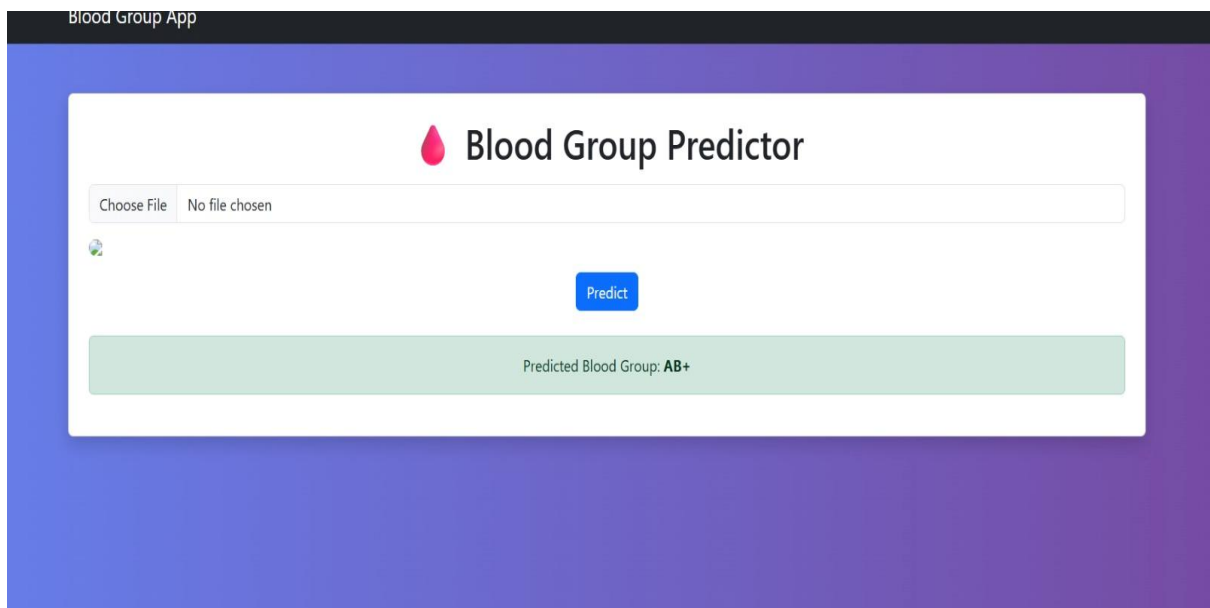
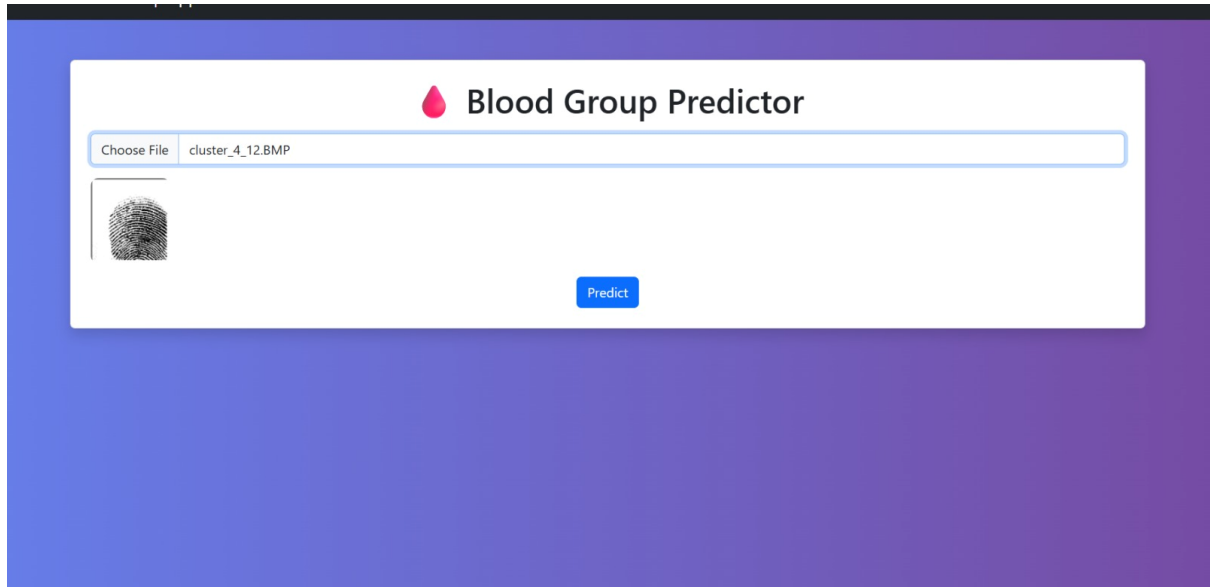
- Result Display Module

OUTPUT SCREENSHOTS (console-based output)

```
PS C:\Finger printing> python hoig1.py
Total images loaded: 6000
K = 1 Accuracy = 0.75
K = 2 Accuracy = 0.75
K = 3 Accuracy = 0.7916666666666666
K = 4 Accuracy = 0.7866666666666666
K = 5 Accuracy = 0.8091666666666667
K = 6 Accuracy = 0.8083333333333333
K = 7 Accuracy = 0.82
K = 8 Accuracy = 0.8275
K = 9 Accuracy = 0.8291666666666667
K = 10 Accuracy = 0.8325
K = 11 Accuracy = 0.8416666666666667
K = 12 Accuracy = 0.8441666666666666
K = 13 Accuracy = 0.85
K = 14 Accuracy = 0.8466666666666667
K = 15 Accuracy = 0.8466666666666667
K = 16 Accuracy = 0.8475
K = 17 Accuracy = 0.8483333333333334
K = 18 Accuracy = 0.8425
K = 19 Accuracy = 0.85
Best K: 13
Model Accuracy: 85.00%
Predicted Blood Group: O+
PS C:\Finger printing> █
```

WEB BASED OUTPUT





CONCLUSION

The proposed system successfully demonstrates that fingerprint patterns can be used to predict blood groups using machine learning techniques. This method provides a fast and non-invasive alternative to traditional blood testing methods. It can be useful in emergency situations where quick identification is required.

FUTURE SCOPE

- Improve accuracy using deep learning models
- Develop mobile application for real-time detection
- Integrate with hospital databases
- Expand dataset for better classification
- Improve fingerprint image processing techniques